



Worksheet W4: Slope — Rise Over Run

"Slope = Speed"

Chapter 4 · 12 Problems · Name: _____ Date: _____

Problem 1

Question: What is **slope** in mathematics?

💡 *Hint: Think about the steepness of a hill.*

Problem 2

Question: What is the **Rule of Two**?

💡 *Hint: Slope is always one thing divided by another.*

Problem 3

Question: Find the slope between the points $(2, 3)$ and $(6, 11)$.

💡 *Hint: $\text{Slope} = \text{Rise} \div \text{Run} = (y_2 - y_1) \div (x_2 - x_1)$.*

Problem 4

Question: Match each slope description:

A. Large positive slope	1. The line is flat
B. Small positive slope	2. The line goes down
C. Zero slope	3. A steep hill going up
D. Negative slope	4. A gentle hill going up

💡 *Hint: Think about walking up hills.*

Problem 5

Question: Calculate the **rise** and **run** for points $(1, 4)$ and $(5, 12)$.

💡 *Hint: Rise = $y_2 - y_1$, Run = $x_2 - x_1$.*

Problem 6

Question: What does a **negative slope** look like on a graph?

💡 *Hint: A line going down from left to right.*

Problem 7

Question: What is the **slope formula** in algebra?

💡 *Hint: $m = (y_2 - y_1) / (x_2 - x_1)$.*

Problem 8

Question: Find the slope between points $(0, 2)$ and $(4, 10)$.

💡 *Hint: Rise = $10 - 2 = 8$, Run = $4 - 0 = 4$.*

Problem 9

Question: What is the **tangent line** to a curve?

💡 *Hint: It touches the curve at exactly one point.*

Problem 10

Question: For $y = x^2$, what is the slope at $x = 3$?

💡 *Hint: The derivative of x^2 is $2x$. Plug in $x = 3$.*

Problem 11

Question: A line has slope 3 and passes through $(1, 5)$. What is y when $x = 4$?

💡 *Hint: $3 = (y - 5) \div (4 - 1)$.*

Problem 12

Question: Why is **slope = speed**?

💡 *Hint: Both involve dividing one change by another.*

